

Faithfulness and Restriction

Every object built before it is used, every result derived rather than quoted

A working companion to the project Variational Quantum Circuits for Pure Spherical Manifolds

Purpose Enough understanding to write the report yourself — the mathematics, the method, and the measurements

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HOW TO READ THIS

Nothing is assumed beyond secondary-school trigonometry and the idea of a vector. Every symbol is defined where it first appears, and every result is derived in full — there are no steps left "to the reader" and no appeals to authority.

The three ideas you have seen named together — **geodesic angle**, **fidelity**, **kernel** — are not three separate topics. They are one object seen from three directions, and §4 derives that identity in six lines. Everything afterwards is consequence.

Green boxes marked *check yourself* contain a short exercise. Do them: they are the difference between recognising a derivation and being able to reproduce it in your own words, which is what writing the report requires.

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Part I — The objects

1. The sphere, and two ways to name a point

Some data have a direction but no meaningful length: the way a compass needle points, where a star sits on the night sky, the orientation frozen into a magnetic mineral, the bearing a bird flew. Strip away the length and what remains is a point on the surface of a ball of radius one. Write that surface S^2 — the "2" is its dimension, because two numbers are enough to name any point on it, even though it sits inside three-dimensional space.

The first way to name a point uses two angles. Let θ (theta) be the angle measured down from the north pole, running from 0 at the pole itself to π at the south pole. Let φ (phi) be the angle measured around the equator, running from 0 to 2π . The pair (θ, φ) is called a **coordinate chart**: a scheme for naming points with numbers. Geographers call these colatitude and longitude.

The second way gives the point's position in ordinary space as three numbers:

$$\mathbf{r} = (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta) \tag{1}$$

Check that this really lies on the sphere by computing its squared length:

$$\begin{aligned} x^2 + y^2 + z^2 &= \sin^2\theta \cos^2\varphi + \sin^2\theta \sin^2\varphi + \cos^2\theta \\ &\quad \text{group the first two terms} \\ &= \sin^2\theta (\cos^2\varphi + \sin^2\varphi) + \cos^2\theta \\ &\quad \cos^2 + \sin^2 = 1, \text{ twice} \\ &= \sin^2\theta + \cos^2\theta = 1 \quad \checkmark \end{aligned}$$

The two descriptions carry the same information, but not in the same way, and the difference matters for everything that follows. The three numbers (x, y, z) describe *where the point is*. The two angles describe *where the point is relative to a pole and a meridian that somebody chose*. Nothing in the data picks out a north pole; that choice is ours.

Two consequences of that arbitrariness will come back repeatedly.

Longitude wraps. The values $\varphi = 0$ and $\varphi = 2\pi$ name the same point, so as a number φ jumps by 2π while the point does not move at all. Any model that reads φ as a plain number sees a cliff where the sphere is smooth.

Longitude collapses at the poles. At $\theta = 0$, equation (1) gives $\mathbf{r} = (0, 0, 1)$ no matter what φ is. Every value of longitude names the *same* point. The chart assigns infinitely many names to one place — a genuine feature of the sphere's shape, not a defect of the formula, and no chart on a sphere can avoid it. This is why a sphere is not a rectangle in disguise.

CHECK YOURSELF

Two points have $(\theta, \varphi) = (0.01, 0)$ and $(0.01, \pi)$. They are both within 0.01 radians of the north pole, so they are close together on the sphere. How far apart are their coordinate pairs, treated as points in a plane? Now compute the same for $(\theta, \varphi) = (\pi/2, 0)$ and $(\pi/2, \pi)$. What does the comparison tell you about using (θ, φ) directly as input to a model?

2. Distance on a sphere: the geodesic angle

Between two points on a sphere, what is "the distance"? The straight line joining them cuts through the interior, which is not a path on the surface. The shortest path *along the surface* is an arc of the great circle through both points — the circle you get by slicing the sphere with a plane through its centre. Such a shortest surface path is called a **geodesic**, and since the radius is one, the length of that arc equals the angle it subtends at the centre.

Call that angle γ (gamma), the **geodesic angle**. The definition of the dot product between two unit vectors gives it immediately:

$$\mathbf{r} \cdot \mathbf{r}' = \cos \gamma, \quad \gamma \in [0, \pi] \quad (2)$$

$\gamma = 0$ means the points coincide, $\gamma = \pi$ means they are antipodal, and $\cos \gamma$ decreases steadily in between, so γ is a genuine measure of separation.

Writing the dot product out in angles gives a formula worth knowing:

$$\begin{aligned} \mathbf{r} \cdot \mathbf{r}' &= \sin \theta \cos \varphi \cdot \sin \theta' \cos \varphi' + \sin \theta \sin \varphi \cdot \sin \theta' \sin \varphi' + \cos \theta \cos \theta' \\ &\quad \text{factor } \sin \theta \sin \theta' \text{ out of the first two terms} \\ &= \sin \theta \sin \theta' (\cos \varphi \cos \varphi' + \sin \varphi \sin \varphi') + \cos \theta \cos \theta' \\ &\quad \text{the bracket is the cosine addition formula} \\ &= \sin \theta \sin \theta' \cos(\varphi - \varphi') + \cos \theta \cos \theta' \quad (3) \end{aligned}$$

Equation (3) is the spherical law of cosines. Notice what it does and does not contain: φ and φ' appear only through their *difference*, because rotating the whole sphere about the polar axis cannot change how far apart two points are.

That observation generalises, and it is the reason γ is the right notion of distance rather than merely one option among many. A **rotation** is a linear map R on three-dimensional space that preserves lengths and angles; the collection of all of them is written $SO(3)$. Rotations preserve dot products, so if we replace every point $\mathbf{r}$ by $R\mathbf{r}$, then

$$(R\mathbf{r}) \cdot (R\mathbf{r}') = \mathbf{r} \cdot \mathbf{r}' = \cos \gamma$$

and γ is unchanged. Every geodesic angle in a dataset survives a rotation untouched. A rotated dataset is, in every respect that the sphere's own geometry can detect, *the same dataset* — only the coordinate names have changed. We will use this later as a measuring instrument, and it is worth pausing on why it is such a good one: it lets us change the coordinates while provably holding the problem fixed.

CHECK YOURSELF

Use (3) to compute γ between $(\theta, \varphi) = (\pi/2, 0)$ and $(\pi/2, \pi/2)$. Then compute it between $(\pi/4, 0)$ and $(\pi/4, \pi/2)$. The change in longitude is $\pi/2$ in both cases — why are the answers different, and which one is larger?

3. The qubit, and why its states form a sphere

A classical bit is 0 or 1. A **qubit** is described by two complex numbers, written

$$|\psi\rangle = a|0\rangle + b|1\rangle, \quad |a|^2 + |b|^2 = 1 \quad (4)$$

The symbols $|0\rangle$ and $|1\rangle$ are just names for the two basic outcomes; the notation $|\cdot\rangle$ is standard in quantum mechanics and carries no meaning beyond "this is a state vector". The numbers a and b are called **amplitudes**. When the qubit is measured it returns outcome 0 with probability $|a|^2$ and outcome 1 with probability $|b|^2$, which is why the two must sum to one.

How many real numbers does it take to specify a qubit? Two complex numbers are four real numbers. The normalisation $|a|^2 + |b|^2 = 1$ removes one. One more is removed by a fact of quantum mechanics: multiplying the whole state by a phase factor $e^{i\mu}$ — a complex number of modulus one — changes no measurable prediction whatsoever, because every probability depends on a modulus squared and $|e^{i\mu}c|^2 = |c|^2$. Such an overall factor is called a **global phase** and is physically invisible. Four minus one minus one leaves **two** real parameters.

Two real parameters, one of them periodic. Let us make that concrete. Since $|a|^2 + |b|^2 = 1$, we may write $|a| = \cos(\theta/2)$ and $|b| = \sin(\theta/2)$ for a unique $\theta \in [0, \pi]$ — the half-angle is a convention that will pay off shortly. Writing the phases of a and b as μ and ν ,

$$\begin{aligned} |\psi\rangle &= e^{i\mu}\cos(\theta/2)|0\rangle + e^{i\nu}\sin(\theta/2)|1\rangle \\ &\quad \text{multiply by the global phase } e^{-i\mu}, \text{ which changes nothing physical} \\ &\simeq \cos(\theta/2)|0\rangle + e^{i(\nu-\mu)}\sin(\theta/2)|1\rangle \\ &\quad \text{name the surviving phase difference } \varphi \\ &= \cos(\theta/2)|0\rangle + e^{i\varphi}\sin(\theta/2)|1\rangle \quad (5) \end{aligned}$$

Every qubit state can be written in the form (5) with $\theta \in [0, \pi]$ and $\varphi \in [0, 2\pi)$. Those are exactly the coordinates of a point on a sphere, from §1. The correspondence is not an illustration or a convenient picture — it is an exact, reversible identification, and the sphere of qubit states is called the **Bloch sphere**.

Note that the collapse of longitude at the pole appears here too, and for the same reason. At $\theta = 0$ we have $\sin(\theta/2) = 0$, so the phase $e^{i\varphi}$ multiplies zero and (5) reduces to $|0\rangle$ regardless of φ . The state space knows that all longitudes at the pole are one point, automatically, with no special handling.

Gates, and the encoding

A **quantum gate** acting on one qubit is a 2×2 matrix U that preserves normalisation — a **unitary** matrix, meaning $U^\dagger U = I$, where $U^\dagger$ is the conjugate transpose. The two we need are rotations about axes of the Bloch sphere:

$$RY(\theta) = \begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix} \quad RZ(\varphi) = \begin{bmatrix} e^{-i\varphi/2} & 0 \\ 0 & e^{i\varphi/2} \end{bmatrix} \quad (6)$$

Encoding means turning a classical data point into a quantum state. Our encoding starts from $|0\rangle$ — which as a column vector is $(1, 0)^T$ — and applies RY then RZ, using the data point's own two angles as the gate parameters:

$$\begin{aligned} \text{RY}(\theta)|0\rangle &= \cos(\theta/2)|0\rangle + \sin(\theta/2)|1\rangle \\ &\text{read off the first column of RY} \\ \text{RZ}(\varphi) \text{RY}(\theta)|0\rangle &= e^{-i\varphi/2}\cos(\theta/2)|0\rangle + e^{i\varphi/2}\sin(\theta/2)|1\rangle \\ &\text{factor out the global phase } e^{-i\varphi/2}, \text{ which is invisible} \\ &\simeq \cos(\theta/2)|0\rangle + e^{i\varphi}\sin(\theta/2)|1\rangle =: |q(\theta, \varphi)\rangle \quad (7) \end{aligned}$$

Compare (7) with (5): they are identical. **The encoding places the data point at exactly its own location on the qubit's sphere.** Nothing is approximated, projected, or flattened.

A TRAP WORTH KNOWING ABOUT

The operator convention is $U = \text{RZ}(\varphi) \text{RY}(\theta)$ — RZ written on the left, because it acts second. Software libraries such as PennyLane specify circuits in the order the gates are *applied*, so the correct code is `RY(theta)` first, then `RZ(phi)`. Circuit order is the reverse of matrix order. Both conventions are standard and they look contradictory side by side, which is exactly why our code carries a test that pins it.

That test contains a second trap. Because RZ contributes the global phase $e^{-i\varphi/2}$, the state produced by the circuit does *not* equal (7) entry by entry — it equals it up to that phase. A test comparing amplitudes directly **fails on a correct implementation**. The right check is $|\langle q_{\text{analytic}} | q_{\text{circuit}} \rangle| = 1$, which is insensitive to global phase.

4. Density matrices, fidelity, and the kernel — one object, three views

This is the section that ties the three named concepts together. We need one more piece of machinery first, and it is worth the detour because it turns hard calculations into easy ones.

The density matrix

For a state $|\psi\rangle$, define the **density matrix** $\rho = |\psi\rangle\langle\psi|$, where $\langle\psi|$ is the conjugate-transpose row vector. For $|\psi\rangle = a|0\rangle + b|1\rangle$ this is the 2×2 matrix

$$\rho = \begin{bmatrix} |a|^2 & ab^* \\ a^*b & |b|^2 \end{bmatrix}$$

where $*$ denotes complex conjugation. The density matrix contains exactly the physically meaningful information and discards the global phase automatically: replacing $|\psi\rangle$ by $e^{i\mu}|\psi\rangle$ leaves ρ unchanged, because the phase cancels against its own conjugate.

Now introduce the three **Pauli matrices**, which are simply a convenient basis for 2×2 matrices:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad (8)$$

Write $\sigma = (X, Y, Z)$, so that for a real 3-vector $\mathbf{r}$ the expression $\mathbf{r} \cdot \sigma$ means $r_1 X + r_2 Y + r_3 Z$. Together with the identity matrix I , these four matrices span all 2×2 matrices, so *every* density matrix can be written as a combination of them. For our states the combination is remarkably clean:

$$\rho = \frac{1}{2} (I + \mathbf{r} \cdot \sigma) \quad (9)$$

where $\mathbf{r}$ is precisely the Bloch vector of §3. Verify it by expanding the right-hand side:

$$\frac{1}{2}(I + x X + y Y + z Z) = \frac{1}{2} \begin{bmatrix} 1+z & x-iy \\ x+iy & 1-z \end{bmatrix}$$

substitute $z = \cos \theta$ and use $1 + \cos \theta = 2\cos^2(\theta/2)$

$$\text{top-left entry: } (1 + \cos \theta)/2 = \cos^2(\theta/2) = |a|^2 \quad \checkmark$$

substitute $x = \sin \theta \cos \phi$, $y = \sin \theta \sin \phi$, and use $\sin \theta = 2 \sin(\theta/2)\cos(\theta/2)$

$$\text{top-right entry: } (x - iy)/2 = \sin \theta e^{-i\phi}/2 = \cos(\theta/2) \sin(\theta/2) e^{-i\phi} = a b^* \quad \checkmark$$

Every entry matches, so (9) holds. **Read what it says: the density matrix is a constant plus something linear in the data point $\mathbf{r}$.** This single structural fact — that ρ is *affine* in $\mathbf{r}$, meaning constant-plus-linear — is the seed of the entire restriction result in §7. It is worth remembering in exactly that form.

Fidelity

Given two states, how similar are they? The natural quantum answer is the **fidelity**: the squared modulus of their overlap,

$$F(\psi, \psi') = |\langle \psi | \psi' \rangle|^2 \quad (10)$$

This is not an arbitrary formula. It is the probability that a measurement designed to detect $|\psi\rangle$ returns "yes" when handed $|\psi'\rangle$ — an operational quantity you could measure in a laboratory. It equals 1 when the states are identical and 0 when they are perfectly distinguishable. In density-matrix language it takes the compact form $F = \text{Tr}[\rho \rho']$, where Tr denotes the trace (the sum of diagonal entries).

The kernel

Independently of any of this, machine learning has the notion of a **kernel**: a function $k(p, p')$ that scores how similar two data points are, with the requirement that it behaves like an inner product. Kernels are the object that methods such as support vector machines and Gaussian processes are built on, because once you have a similarity score you can do a great deal of learning without ever writing down explicit features.

The fidelity of two encoded states is a kernel on the original data — it takes two data points, encodes them, and returns a similarity score. Now we compute it, and everything collapses to one line.

$$k(p, p') = |\langle q(p) | q(p') \rangle|^2 = \text{Tr}[\rho \rho']$$

substitute (9) for both density matrices

$$= \frac{1}{4} \text{Tr}[(I + \mathbf{r} \cdot \sigma)(I + \mathbf{r}' \cdot \sigma)]$$

expand the product into four terms

$$= \frac{1}{4} (\text{Tr}[I] + \text{Tr}[\mathbf{r}' \cdot \sigma] + \text{Tr}[\mathbf{r} \cdot \sigma] + \text{Tr}[(\mathbf{r} \cdot \sigma)(\mathbf{r}' \cdot \sigma)])$$

$\text{Tr}[I] = 2$; each Pauli matrix is traceless, so the two middle terms vanish

$$\begin{aligned}
& \text{and } \text{Tr}[(\mathbf{a} \cdot \boldsymbol{\sigma})(\mathbf{b} \cdot \boldsymbol{\sigma})] = 2 \mathbf{a} \cdot \mathbf{b}, \text{ from } \sigma_i \sigma_j = \delta_{ij} I + i \epsilon_{ijk} \sigma_k \\
& = \frac{1}{4} (2 + 0 + 0 + 2 \mathbf{r} \cdot \mathbf{r}') = (1 + \mathbf{r} \cdot \mathbf{r}')/2 \\
& \text{and } \mathbf{r} \cdot \mathbf{r}' = \cos \gamma, \text{ by equation (2)}
\end{aligned}$$

$$\mathbf{k}(\gamma) = (1 + \cos \gamma) / 2 = \cos^2(\gamma/2) \quad (11)$$

There is the connection you were looking for, stated plainly. **The geodesic angle is the sphere's own notion of distance. The fidelity is the quantum notion of similarity. The kernel is the machine-learning notion of similarity. Equation (11) says all three are the same object.** The circuit's measure of how alike two inputs are is the sphere's measure of how far apart they are, with no modelling choice in between.

The final step of (11) uses the half-angle identity $\cos^2(\gamma/2) = (1 + \cos \gamma)/2$, and that form makes two things obvious: k is never negative, and it falls steadily from 1 at $\gamma = 0$ to 0 at $\gamma = \pi$.

CHECK YOURSELF

Reproduce the derivation of (11) on paper without looking, given only $\rho = \frac{1}{2}(I + \mathbf{r} \cdot \boldsymbol{\sigma})$, $\text{Tr}[I] = 2$, $\text{Tr}[\sigma_i] = 0$, and $\text{Tr}[(\mathbf{a} \cdot \boldsymbol{\sigma})(\mathbf{b} \cdot \boldsymbol{\sigma})] = 2\mathbf{a} \cdot \mathbf{b}$. It is four lines. If you can do this, you can derive the central claim of the project from scratch — it really is that short, and saying so in the report is a strength rather than a modesty.

Part II — The two properties

5. Faithfulness

Definition. An encoding is **faithful** if its induced kernel is a strictly decreasing function of the geodesic angle γ , and of nothing else.

Equation (11) shows ours is: k depends on γ alone, and $\cos^2(\gamma/2)$ decreases strictly on $[0, \pi]$. Two consequences follow, and both are worth stating carefully because they are what the word is doing.

First: the encoding is blind to the coordinate frame. Rotations preserve γ (§2), so rotating the coordinate system — moving the north pole, moving the prime meridian — changes no kernel value anywhere. The representation describes *points*, not *coordinates*. Where you chose to put the pole is information the model never receives.

Second: the pole is handled correctly for free. At $\theta = 0$ the state is $|0\rangle$ for every φ (§3), so all longitudes at the pole map to one state, exactly as they name one point. The identification is enforced by the algebra, with no special-case code.

An unfaithful representation, for contrast

The standard classical way to feed angles to a network is to supply their sines and cosines, $(\sin \theta, \cos \theta, \sin \varphi, \cos \varphi)$, which removes the wrap-around discontinuity of §1. Its natural kernel is the dot product of those feature vectors:

$$\begin{aligned} k_{\text{sincos}} &= \sin \theta \sin \theta' + \cos \theta \cos \theta' + \sin \varphi \sin \varphi' + \cos \varphi \cos \varphi' \\ &\quad \text{the cosine addition formula, applied twice} \\ &= \cos(\theta - \theta') + \cos(\varphi - \varphi') \quad (12) \end{aligned}$$

Compare (12) with (11). Ours depends on γ ; this one depends on the two coordinate *differences* separately. Those are not the same thing, and the difference is not subtle. Two pairs of points at identical geodesic separation can receive different similarity scores, and moving the pole changes the answer.

The structural diagnosis is that (12) treats θ and φ as two independent circles. The product of two circles is a **torus** — a doughnut surface — not a sphere. In particular a torus has no poles: on a torus, the "longitude" circle never collapses to a point, so the representation has no way to express the fact that at $\theta = 0$ all values of φ mean the same place. It is not that the sine–cosine features are badly chosen; they faithfully represent the wrong shape.

CHECK YOURSELF

Take the pair $(\theta, \varphi) = (0.1, 0)$ and $(0.1, \pi)$, and the pair $(\pi/2, 0)$ and $(\pi/2, 0.2)$. Use (3) to confirm the two pairs have nearly the same γ . Then evaluate (11) and (12) on both pairs. Which kernel gives the same answer for both, and which does not? This single numerical comparison is the whole content of "unfaithful", and a version of it is asserted as a test in the code.

6. Functions on a sphere: degree, harmonics, Legendre

Restriction is a statement about *which functions on the sphere the model can produce*, so we need a way to classify such functions. The classification is standard and completely analogous to decomposing a sound into pitches, except that the "pitches" live on a sphere.

The building blocks are the **spherical harmonics**, written $Y_{\ell m}(\theta, \phi)$. They are labelled by two integers: a **degree** $\ell = 0, 1, 2, \dots$ and an **order** m running from $-\ell$ to $+\ell$, so there are exactly $2\ell + 1$ harmonics of degree ℓ . What matters for us is what each degree contains:

Degree $\ell = 0$ — one function

The constant. It cannot distinguish any two points.

Degree $\ell = 1$ — three functions

Spanning exactly the three coordinate functions x, y and z . A function of degree 1 is a linear combination of the coordinates: it varies smoothly across the sphere with a single "high side" and a single "low side".

Degree $\ell = 2$ — five functions

Spanning the quadratic combinations, such as $3z^2 - 1$ and xy . These can have two separate high regions — for instance, large near *both* poles and small around the equator.

Higher ℓ

Progressively finer variation, more alternations across the surface.

Every reasonable function on the sphere is a sum of these, and the set of degrees that appear tells you how intricate the function can be. Degree is the sphere's analogue of frequency.

The harmonics are **orthonormal**: integrating the product of two different ones over the sphere gives zero, and each squared integrates to one. That is what makes the decomposition well defined, and it is also what lets us compute it numerically — see §12.

Functions of γ alone

Our kernel is special: it depends only on the angle between two points, not on where either of them sits. For such functions the decomposition simplifies to a one-dimensional one, using the **Legendre polynomials** $P_\ell(t)$. They are the natural basis for functions of a single variable $t \in [-1, 1]$, and the first few are

$$P_0(t) = 1, \quad P_1(t) = t, \quad P_2(t) = (3t^2 - 1)/2, \quad P_3(t) = (5t^3 - 3t)/2 \quad (13)$$

Writing $t = \cos \gamma$, any kernel that depends on γ alone can be expanded as $k(\gamma) = \sum_\ell a_\ell P_\ell(\cos \gamma)$. The connection between the two bases is the **addition theorem**: $P_\ell(\cos \gamma)$ is, up to a constant, the sum over m of $Y_{\ell m}(p) Y_{\ell m}(p')$.

In words: *Legendre degree ℓ and harmonic degree ℓ are the same degree*. A kernel whose Legendre expansion stops at $\ell = 1$ produces only functions whose harmonic expansion stops at $\ell = 1$.

Two results we will use

Schoenberg's theorem tells us which functions of γ are legitimate kernels on a sphere — meaning they behave like inner products, which is what kernel methods require. The condition is beautifully simple: the Legendre coefficients must all be non-negative, $a_\ell \geq 0$ for every ℓ . Nothing else is needed.

The Funk–Hecke formula gives the eigenvalues of the kernel. Writing the kernel as a function f of $t = \cos \gamma$, the eigenvalue attached to degree ℓ is

$$\lambda_\ell = \frac{1}{2} \int_{-1}^1 f(t) P_\ell(t) dt \quad (14)$$

and every harmonic of degree ℓ shares that same eigenvalue — all $2\ell + 1$ of them. That shared value is called a **degeneracy**, and it will explain a pattern in the numbers shortly. Eigenvalues matter because they say how much "room" the kernel gives to each degree: a degree with eigenvalue zero is a degree the model cannot use at all.

7. Restriction, derived twice

We now derive the same conclusion by two independent routes. Agreement between them is the strongest evidence available that the result is real rather than an artefact of one way of looking.

Route one: expand the kernel

Take (11) and read it against (13). Since $P_0 = 1$ and $P_1(t) = t$,

$$k(\gamma) = \frac{1}{2} + \frac{1}{2} \cos \gamma = \frac{1}{2} P_0(\cos \gamma) + \frac{1}{2} P_1(\cos \gamma) \quad (15)$$

so $a_0 = a_1 = \frac{1}{2}$ and $a_\ell = 0$ for every $\ell \geq 2$. The expansion terminates. That termination is what the word **restriction** names.

Two side observations. Both coefficients are non-negative, so Schoenberg's condition is satisfied and k is a legitimate geodesic kernel — this is not a formality, since (12) fails it. And the eigenvalues follow from (14):

$$\lambda_0 = \frac{1}{2} \int_{-1}^1 (1+t)/2 \cdot 1 dt = \frac{1}{4} [t + t^2/2]_{-1}^1 = \frac{1}{4} (1.5 - (-0.5)) = \mathbf{1/2}$$

$$\lambda_1 = \frac{1}{2} \int_{-1}^1 (1+t)/2 \cdot t dt = \frac{1}{4} [t^2/2 + t^3/3]_{-1}^1 = \frac{1}{4} (5/6 - 1/6) = \mathbf{1/6}$$

$$\lambda_2 = \frac{1}{2} \int_{-1}^1 (1+t)/2 \cdot (3t^2-1)/2 dt = \frac{1}{8} \int_{-1}^1 (3t^2 - 1 + 3t^3 - t) dt = \mathbf{0}$$

the odd powers integrate to zero over a symmetric interval, and $\int 3t^2 dt = 2$ cancels $\int -1 dt = -2$

Route two: look at the circuit

This route explains something the first does not: *why depth is irrelevant*.

A variational quantum circuit does three things. It encodes the data point, giving ρ . It applies a unitary $U(w)$ built from trainable gate angles w , possibly involving extra qubits. It measures an observable O , returning the number $f = \text{Tr}[U(\rho \otimes \rho_{\text{anc}}) U^\dagger O]$, where ρ_{anc} is the fixed starting state of any ancillary qubits.

Now define the map Λ that takes a density matrix to that final number:

$$\Lambda(\rho) = \text{Tr}[U(\rho \otimes \rho_{\text{anc}}) U^\dagger O]$$

Λ is linear in ρ . Every operation composing it — tensoring with a fixed state, conjugating by a fixed unitary, tracing against a fixed observable — is linear, and a composition of linear maps is linear. This holds however deep U is, however many ancillas are present, and whatever O is measured.

Substituting (9) and using linearity:

$$f = \Lambda(\rho) = \Lambda\left(\frac{1}{2}I + \frac{1}{2} \sum_i r_i \sigma_i\right)$$

linearity lets the sum and the constants pass through Λ

$$= \frac{1}{2}\Lambda(I) + \frac{1}{2} \sum_i r_i \Lambda(\sigma_i)$$

name $\alpha = \frac{1}{2}\Lambda(I)$ and $\beta_i = \frac{1}{2}\Lambda(\sigma_i)$; these are fixed numbers once the weights are fixed

$$\mathbf{f}(\boldsymbol{\theta}, \boldsymbol{\varphi}) = \boldsymbol{\alpha} + \boldsymbol{\beta} \cdot \mathbf{r} \quad (16)$$

The circuit's output is an affine function of the data point — **four real numbers**, α and the three components of $\boldsymbol{\beta}$, no matter what else is going on inside. Comparing with §6, $\alpha + \boldsymbol{\beta} \cdot \mathbf{r}$ is a combination of the constant and the three degree-1 harmonics, so f has degree $\ell \leq 1$. The two routes agree.

WHAT THIS COSTS THE MODEL

Consider the set of points the model labels positive, $\{p : f(p) > 0\}$. By (16) this is $\{\mathbf{r} : \boldsymbol{\beta} \cdot \mathbf{r} > -\alpha\}$, which is the part of the sphere on one side of a plane — a **spherical cap**, bounded by a single circle.

So a single-encoding variational circuit of this type can draw **exactly one circular decision boundary on the sphere, and nothing else**. Adding layers adds parameters and adds no expressive power whatsoever. The cap on what the model can express lies entirely in the encoding; the trainable part contributes nothing to it.

Say the two properties side by side, because the report will need this sentence: **faithfulness is which basis the expansion is written in; restriction is how many terms it has**. Both originate in the encoding. Neither is affected by the ansatz.

8. The Gram matrix has rank four

Here is a third, entirely elementary route to the same fact, and it is the one that is easiest to check numerically.

Given n data points, the **Gram matrix** K is the $n \times n$ table of kernel values, $K_{ij} = k(y_{ij})$. Using (11) and writing $\mathbf{1}$ for the column of n ones, and R for the $n \times 3$ matrix whose rows are the points' Cartesian coordinates:

$$K_{ij} = \frac{1}{2} + \frac{1}{2} \mathbf{r}_i \cdot \mathbf{r}_j \implies K = \frac{1}{2} \mathbf{1}\mathbf{1}^T + \frac{1}{2} R R^T \quad (17)$$

The rank of a matrix is the number of independent directions it acts on, and the rank of a sum is at most the sum of the ranks. Here $\mathbf{1}\mathbf{1}^T$ has rank 1, and $R R^T$ has rank at most 3 because R has only three columns. Therefore

$$\text{rank } K \leq 1 + 3 = 4 \quad (18)$$

no matter how many data points there are. A thousand points still give a rank-4 matrix. This is restriction restated as a statement of linear algebra, and it needs no harmonics at all.

The eigenvalues are just as easy to predict. For points spread evenly over the sphere the average of $\mathbf{r}$ is close to zero, so $\mathbf{1}$ is nearly orthogonal to the columns of R , and the two pieces of (17) act on separate directions. The first contributes one eigenvalue of $\frac{1}{2}n$. For the second, even spreading gives $R^T R \approx (n/3)I$, so $R R^T$ has three eigenvalues near $n/3$, each halved to $n/6$. Predicted spectrum:

$$n \cdot (\tfrac{1}{2}, \tfrac{1}{6}, \tfrac{1}{6}, \tfrac{1}{6}), \text{ everything else zero} \quad (19)$$

Note the ratio is **3 : 1 : 1 : 1**, not "3 : 1". The threefold repetition is the m-degeneracy of degree 1 from §6 — three harmonics x, y, z sharing one eigenvalue — and writing "3 : 1" silently hides two of the four numbers. Note also the consistency check: the eigenvalues sum to $n(\frac{1}{2} + 3 \cdot \frac{1}{6}) = n$, and the sum of eigenvalues equals the trace, which here is $\sum_i k(0) = n$ since every diagonal entry is $k(\gamma=0) = 1$. Two independent routes to the same n .

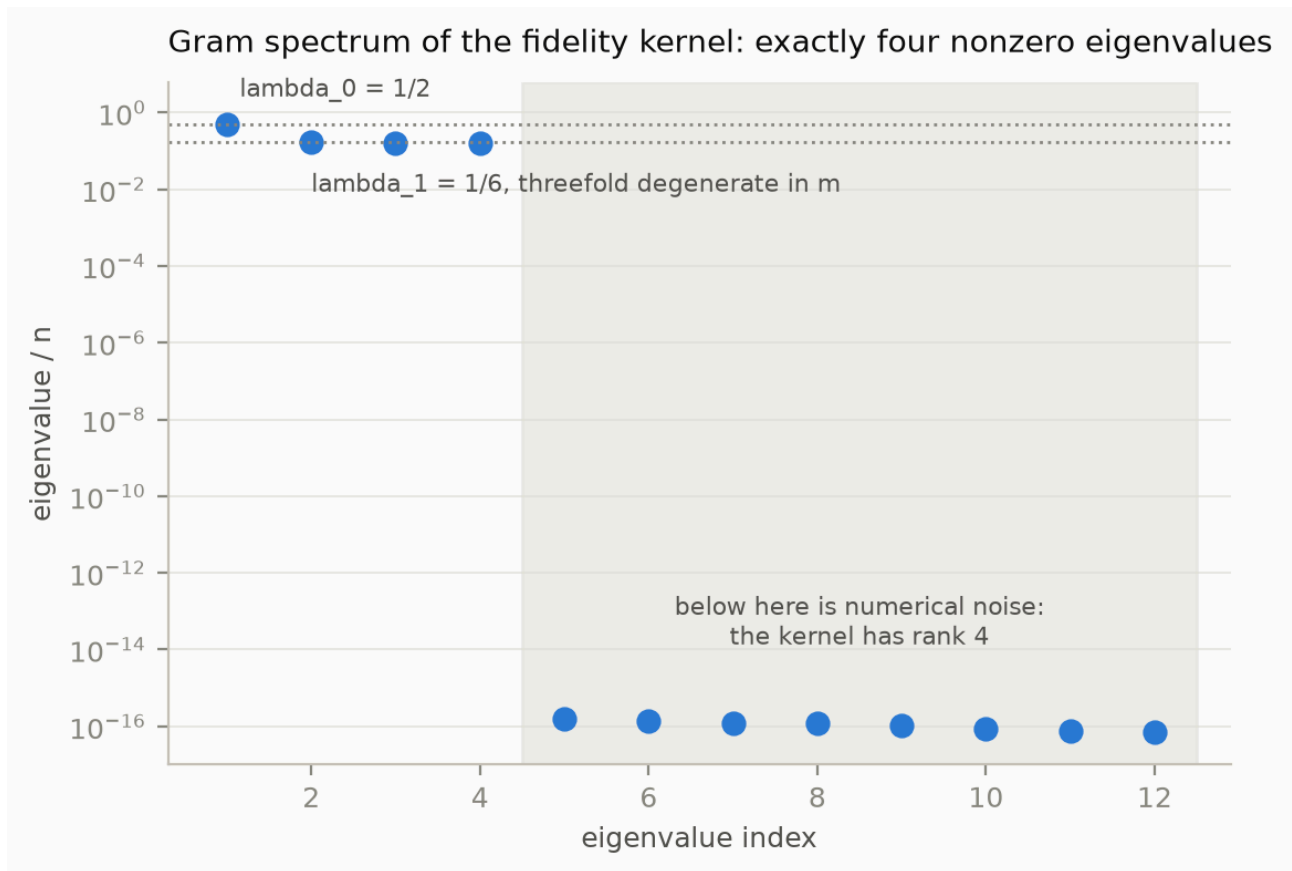


Figure 1. The measured spectrum, 600 points, on a logarithmic vertical axis. Four eigenvalues sit at the predicted heights 0.500, 0.176, 0.162, 0.161 — the last three being the degenerate degree-1 triple — and everything beyond the fourth drops by roughly fourteen orders of magnitude into floating-point rounding error. The cliff is equation (18) made visible.

CHECK YOURSELF

Equation (17) writes K as a sum of two pieces. Which piece corresponds to degree 0, and which to degree 1? Using that, explain without any computation why the sine–cosine kernel (12) also has rank 4 — and then explain why that fact says nothing good about it. (Hint: what four features is (12) an inner product of, and what surface do they parameterise?)

Part III — Raising expressivity

Restriction says the one-upload circuit is limited to degree 1. There are two standard ways to give it more expressive power. They are usually spoken of as interchangeable. They are not, and establishing that is one of the project's genuine contributions.

9. The spin ladder: the same data on n qubits

Apply the same encoding to n qubits at once, feeding every wire the same (θ, φ) . The resulting state is the n -fold tensor product

$$|Q_n(\theta, \varphi)\rangle = |q(\theta, \varphi)\rangle^{\otimes n} \quad (20)$$

This object has a name in physics — the spin- $n/2$ coherent state — but we need only its kernel, and that follows from a single property of tensor products: the overlap of two products is the product of the overlaps.

$$k_n(\gamma) = |\langle Q_n | Q_n' \rangle|^2 = |\langle q | q' \rangle^n|^2 = (|\langle q | q' \rangle|^2)^n$$

the inner bracket is our original kernel (11)

$$= ((1 + \cos \gamma)/2)^n = \cos^{2n}(\gamma/2) \quad (21)$$

Two things follow, and they pull in opposite directions in exactly the way one wants.

Faithfulness survives untouched. k_n is still a function of γ alone, and $\cos^{2n}(\gamma/2)$ is still strictly decreasing on $[0, \pi]$. Every rung of this ladder is faithful.

Restriction relaxes in a controlled way. Writing $t = \cos \gamma$, equation (21) is $((1 + t)/2)^n$, a polynomial in t of degree exactly n . A polynomial of degree n expands in Legendre polynomials with nothing above P_n , so the expansion terminates at $\ell = n$. Turning the dial from n to $n+1$ buys exactly one more degree.

The circuit-side argument of §7 generalises identically. The encoded state is now $\rho^{\otimes n}$, which by (9) is an n -fold product of terms affine in $\mathbf{r}$, hence a polynomial in $\mathbf{r}$ of degree n . Everything downstream is linear, so the output is a degree- n polynomial in the coordinates: $\ell \leq n$, exactly.

The eigenvalues can be computed once and for all from (14), giving

$$\lambda_\ell^{(n)} = (n!)^2 / [(n - \ell)! (n + \ell + 1)!] \quad \text{for } 0 \leq \ell \leq n, \text{ and zero above} \quad (22)$$

Sanity-check it at $n = 1$: for $\ell = 0$ this gives $1/(1! \cdot 2!) = 1/2$, and for $\ell = 1$ it gives $1/(0! \cdot 3!) = 1/6$, reproducing the values computed directly in §7. A useful identity holds at every n : $\sum_\ell (2\ell + 1) \lambda_\ell^{(n)} = 1 = k_n(0)$, where the factor $2\ell + 1$ counts the degenerate harmonics at each degree.

One more observation, useful for the report's discussion. For small γ , $\cos^{2n}(\gamma/2) \approx \exp(-n\gamma^2/4)$, so the kernel sharpens as $1/\sqrt{n}$. This ladder is a tunable-bandwidth geodesic kernel — a knob that makes the similarity measure more discriminating while keeping it geometric.

10. Data re-uploading, and why it is a different knob

The other standard technique feeds the data in several times, with trainable layers in between:

$$E(x) A(w_0) E(x) A(w_1) \dots E(x) A(w_{U-1}) \quad (23)$$

for U **uploads**. The argument for why this increases expressivity is a Fourier argument. Each $RY(\theta)$ or $RZ(\phi)$ contributes terms oscillating at frequency 1 in its angle; passing the data in U times allows products of U such terms, so the output is a trigonometric polynomial in θ and ϕ with frequencies up to U in each. That much is correct and is a well-established result.

The step that does *not* follow is the one that is usually taken next: that bounding the frequency in the coordinates is the same as bounding the harmonic degree on the sphere. It is not, and here is a counterexample the circuit can actually produce at $U = 2$.

Consider the function $\sin \theta \cdot \cos 2\phi$. It has frequency 1 in θ and 2 in ϕ , so it is within reach at two uploads. Is it band-limited in degree? Express it in Cartesian coordinates. From $\cos 2\phi = \cos^2\phi - \sin^2\phi$ and $x = \sin \theta \cos \phi$, $y = \sin \theta \sin \phi$ we get $\cos 2\phi = (x^2 - y^2)/\sin^2\theta$, so

$$\sin \theta \cdot \cos 2\phi = (x^2 - y^2) / \sin \theta = (x^2 - y^2) / \sqrt{1 - z^2} \quad (24)$$

That is not a polynomial in (x, y, z) — the square root in the denominator cannot be removed. Only polynomials have terminating harmonic expansions, so this function has contributions at arbitrarily high degree ℓ . A model that can produce it is not degree-limited at any finite ℓ .

Why do the two notions agree at one upload, then? Because at $U = 1$ the accessible functions are spanned by $\{1, \cos \theta, \sin \theta \cos \phi, \sin \theta \sin \phi\} = \{1, z, x, y\}$, which is precisely the set of harmonics with $\ell \leq 1$. The two descriptions happen to coincide there, which is why the degree-1 result is exact. They part company at $U = 2$.

THE CONTRAST, STATED FOR THE REPORT

	spin ladder (broadcast)	data re-uploading
what it raises	harmonic degree on the sphere	Fourier frequency in the coordinates
rotation-covariant	yes	no
induced feature map	a fixed geodesic kernel, $\cos^{2n}(\gamma/2)$	trainable; not a fixed kernel at all
Schoenberg-admissible	at every rung	not certified at any rung
band-limited on S^2	exactly, $\ell \leq n$	no

One ladder raises degree while staying geometric. The other raises coordinate frequency and leaves the geometric description altogether. They are not two settings of one dial, and predictions of the form "degree ℓ becomes learnable at rung ℓ " hold only for the first.

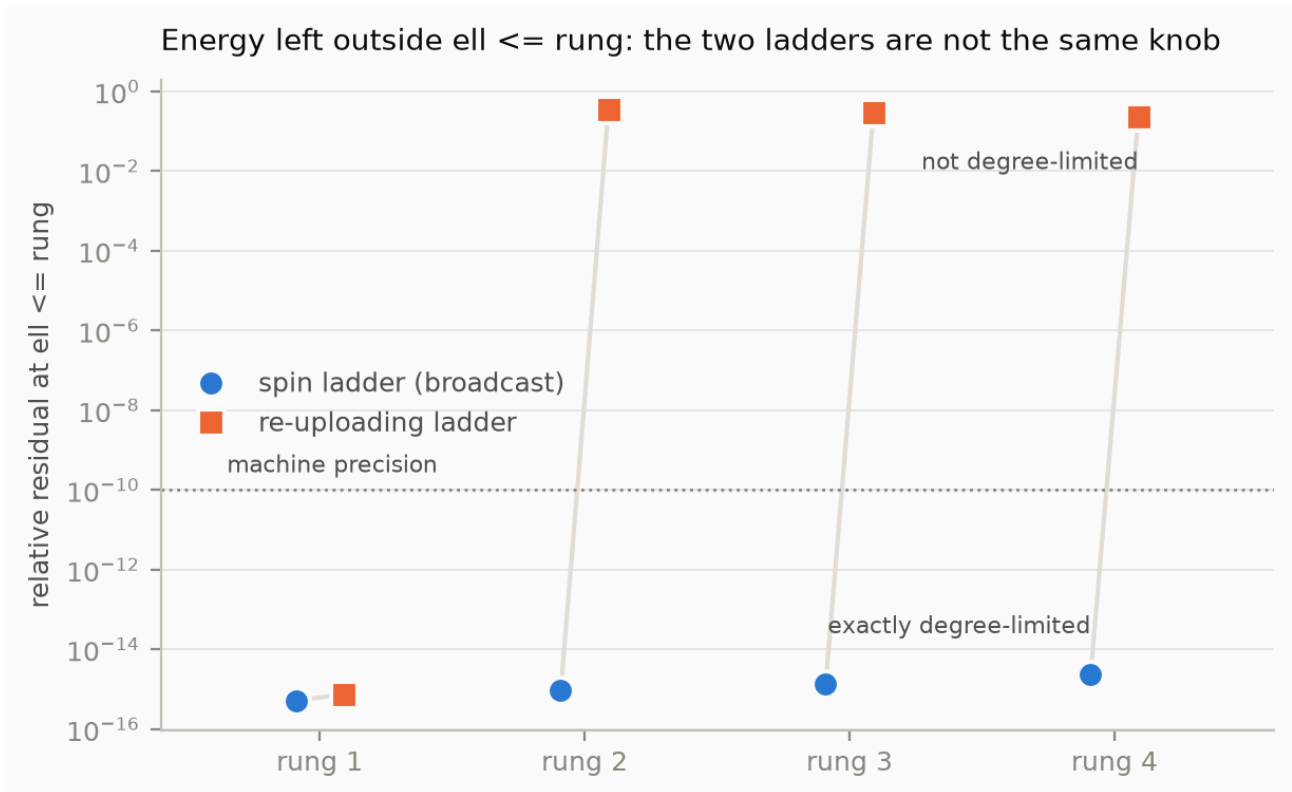


Figure 2. Measured energy left outside each rung's own degree, on a logarithmic axis. The broadcast ladder sits at floating-point zero at every rung — exactly degree-limited, as §9 proves. Re-uploading sits fourteen orders of magnitude above it from rung 2 onward, as §10 predicts. At rung 1 the two coincide, because one upload of one qubit is the same circuit either way.

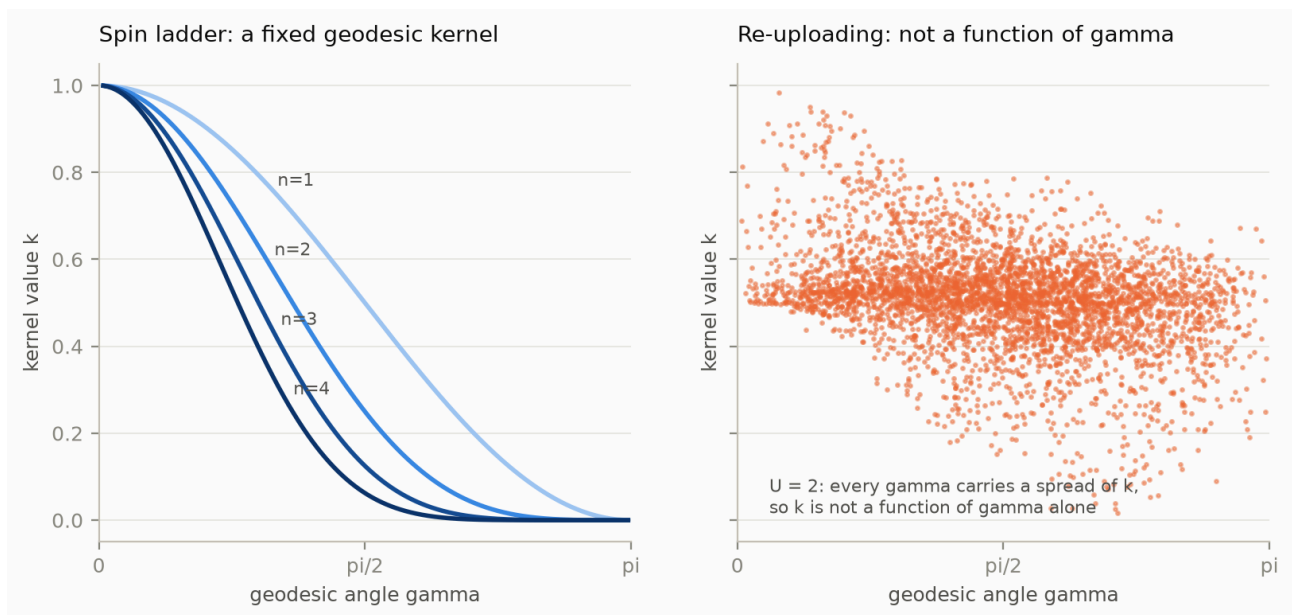


Figure 3. Similarity against geodesic separation. **Left:** each rung of the broadcast ladder collapses onto a single clean curve — the similarity depends on γ and nothing else, which is faithfulness drawn rather than asserted, and the curves sharpen with n exactly as the $1/n$ bandwidth argument predicts. **Right:** a two-upload model assigns a whole spread of similarity values to each separation, because its similarity depends on the coordinates as well. The vertical spread is the finding, not noise.

Part IV — The instruments

A recurring difficulty in this field is that a model failing to learn something admits two explanations: it cannot express the function, or the training procedure did not find it. An accuracy number cannot tell them apart. Three instruments were built to sidestep the ambiguity entirely, and none of them trains anything.

11. The affine ceiling

By (16), the one-upload model must have the form $f = \alpha + \boldsymbol{\beta} \cdot \mathbf{r}$, and by the box that follows it, its positive region is a spherical cap. So the question "what is the best this model could possibly do on this dataset?" has a computable answer, found by searching over caps rather than by training.

A cap is determined by two things: the direction $\boldsymbol{\beta}$ its axis points, and the threshold c where the plane cuts. Fix the direction. Then every point has a scalar projection $s_i = \mathbf{r}_i \cdot \boldsymbol{\beta}$, the classifier is "predict 1 when $s > c$ ", and the best c is found exactly by sorting the projections and scanning: as the threshold sweeps past each point, the accuracy changes by a known amount, so a single pass over the sorted list gives the accuracy at every possible threshold. Take the best, and also the best for the reversed inequality, since "predict 1 below the cut" is an equally valid cap.

Then repeat over many directions, spread evenly using a Fibonacci construction — even spreading matters, because a latitude–longitude grid over-samples near the poles and would bias the result for pole-centred targets.

$$\text{cost: } O(G \cdot n \log n) \text{ for } G \text{ directions and } n \text{ points} \quad (25)$$

The result is an **upper bound on the accuracy of every circuit of this class**, obtained with no optimiser, no random seed and no learning rate. That is what makes it decisive: a number obtained this way cannot be dismissed as an optimisation failure.

ONE CAVEAT TO STATE IN THE REPORT

The direction grid is finite, so the returned number is a *lower* bound on the true ceiling: the best available direction can sit up to half a grid spacing away from the optimum. At 1024 directions the spacing is roughly 6° , costing about one point of accuracy on a sharply defined cap. Use a fine grid for reported numbers, and read a measured accuracy marginally above a coarse-grid ceiling as a resolution artefact rather than a falsification. This matters because exceeding the ceiling is the project's falsification criterion, and one should not falsify a correct theory with a grid artefact.

Worked example: the quadrupole ceiling by hand

This is worth doing by hand, because it converts the instrument from a black box into arithmetic you can put in the report.

The target labels a point positive when $3\cos^2\theta - 1 > 0$, that is when $|z| > 1/\sqrt{3} \approx 0.5774$ — two caps, one around each pole. Points are sampled uniformly on the sphere, which means z is uniform on $[-1, 1]$, a fact worth pausing on: uniform on the sphere is uniform in z , not uniform in θ .

Now find the best single cap. By the target's symmetry about the z -axis the best axis is $\boldsymbol{\beta} = \hat{\mathbf{z}}$, so the classifier is "predict positive when $z > c$ ". Take $c = 0.5774$. Then:

correctly positive: $P(z > 0.5774) = (1 - 0.5774)/2 = 0.2113$

correctly negative: $P(|z| < 0.5774) = 0.5774$

and the region $z < -0.5774$, which is truly positive, is misclassified

total accuracy = $0.2113 + 0.5774 = \mathbf{0.7887}$

The numerical scan returns 0.784 at $n = 20000$, agreeing within sampling noise. Two consequences follow, and both belong in the report.

First, the research plan predicted the circuit would *fail* on this target at 50–65%. It does not: it saturates at 79%. A circuit scoring 79% — the theory working exactly as derived — would have been recorded as the theory failing.

Second, the target's class balance is 42/58, so a model that always guesses the majority class scores 0.575. The predicted 50–65% band straddles that number, meaning it could not have distinguished a genuine expressivity limit from a model that does nothing at all. This is why the majority baseline appears in every table.

12. Harmonic projection

The second instrument asks the circuit directly: *how much of your output lives at each degree?*

Evaluate the trained circuit at a few thousand points spread over the sphere, giving a vector v of outputs. Build the matrix A whose columns are the harmonics $Y_{\ell m}$ evaluated at those same points, for all ℓ up to some cut-off. Solve the least-squares problem $A c \approx v$. The coefficient vector c then splits by degree, and the squared coefficients at each degree give the energy there. Normalising by the total gives a spectrum that sums to one.

The companion quantity is the **residual**: fit using only degrees up to L , and measure what is left over,

$$\text{residual}(L) = \|v - A_{\leq L} c\| / \|v - \text{mean}(v)\| \quad (26)$$

A residual of order 10^{-15} means the function lives entirely within degree L , to the limit of floating-point arithmetic. A residual of order 0.1 means a substantial fraction of the function lies outside.

Note what this does *not* involve: no dataset, no labels, no optimiser, no notion of accuracy. It is a direct measurement of the model's expressive content, which is why it settles the ladder question that accuracy comparisons could only hint at.

One implementation note worth keeping: the harmonics are built from the associated Legendre function `lpmv` rather than from SciPy's `sph_harm`, because that function has changed name and argument order across recent SciPy releases, and a silent argument-order change in a numerical instrument is a very expensive kind of bug.

13. Coordinate leakage, and the dataset gate

The third instrument is about datasets rather than models, and it exists because of a failure found in this project's own earlier work.

For a candidate dataset, compute three numbers: the best accuracy from a single threshold on θ ; the best accuracy from the best *arc* in ϕ (an arc rather than a threshold, because ϕ is circular and a plain threshold would understate what is readable from it); and the affine ceiling. The gap between the ceiling and the better single coordinate is how much the second coordinate actually buys.

If one coordinate alone reaches the ceiling, the task is solvable by reading one number, and nothing can be concluded from it about geometry, topology, or inductive bias.

THE FAILURE THIS CAUGHT

The project's original dataset, `make_sphere_moons`, draws ϕ from $(0, \pi)$ for one class and $(\pi, 2\pi)$ for the other. **Longitude alone classifies it perfectly**: measured 1.000 from ϕ alone, with an affine ceiling of 1.000. Every model reaches 100%.

This is the saturated-benchmark failure mode, and it is the single largest risk the research plan identifies. A task everybody solves measures nothing. Any earlier conclusion about geometric advantage drawn from that dataset was really a conclusion about the ϕ split.

The generator was kept unchanged rather than repaired — it is the dataset behind the existing published figure, and as a documented negative example it is worth more than it would be deleted — but it is now declared as leaking, the experiment code refuses it, and a replacement was built.

The replacement, and a small theorem about it

What does it take for a task to genuinely need both coordinates? Recall from §7 that any task the circuit can solve perfectly is a spherical cap. Rotate the frame so the cap's axis is $\hat{z}$, and the label reads " θ less than some value" — ϕ never appears. So:

A CONSEQUENCE WORTH PUTTING IN THE REPORT

For any task the one-upload circuit can solve, there *exists* a coordinate frame in which one coordinate suffices. Needing both θ and ϕ is therefore a property of the frame you happened to choose, not of the task. That is faithfulness restated from the data's side — and it means "uses both coordinates" and "solvable by the circuit" can coexist only in the sense that the frame is generic.

A task that needs both coordinates in *every* frame must not be a cap, hence must have degree ≥ 2 , hence is provably out of the circuit's reach. The dataset `make_hyperbolic`, labelling by $\text{sign}(x \cdot z)$, is exactly this: both coordinate marginals sit at chance (measured 0.510 and 0.515), and the price is that $x \cdot z \propto \sin \theta \cos \theta \cos \phi$ is a degree-2 harmonic the circuit cannot express.

The practical replacement is `make_tilted_bands`: generate two latitude bands, then rigidly rotate them so the separating circle is not a line of latitude. The boundary becomes $\sin \tau \cdot \sin \theta \cdot \cos(\phi - \psi) + \cos \tau \cdot \cos \theta = c$, which no single threshold on either coordinate reproduces, while the task itself — difficulty, class balance, ceiling — is provably unchanged, because rotations preserve all geodesic distances. Measured at tilt $\tau = \pi/4$: θ alone gives 0.619, ϕ alone 0.611, the ceiling is 0.886. The second coordinate buys 0.267.

Both extremes of the tilt are traps, and knowing why is useful: at $\tau = 0$ the label is readable from θ (0.885), and at $\tau = \pi/2$ the cut passes through the poles and it becomes readable from ϕ (0.884). The leak moves rather than disappearing. This is why the gate reports the numbers rather than trusting a construction to be safe.

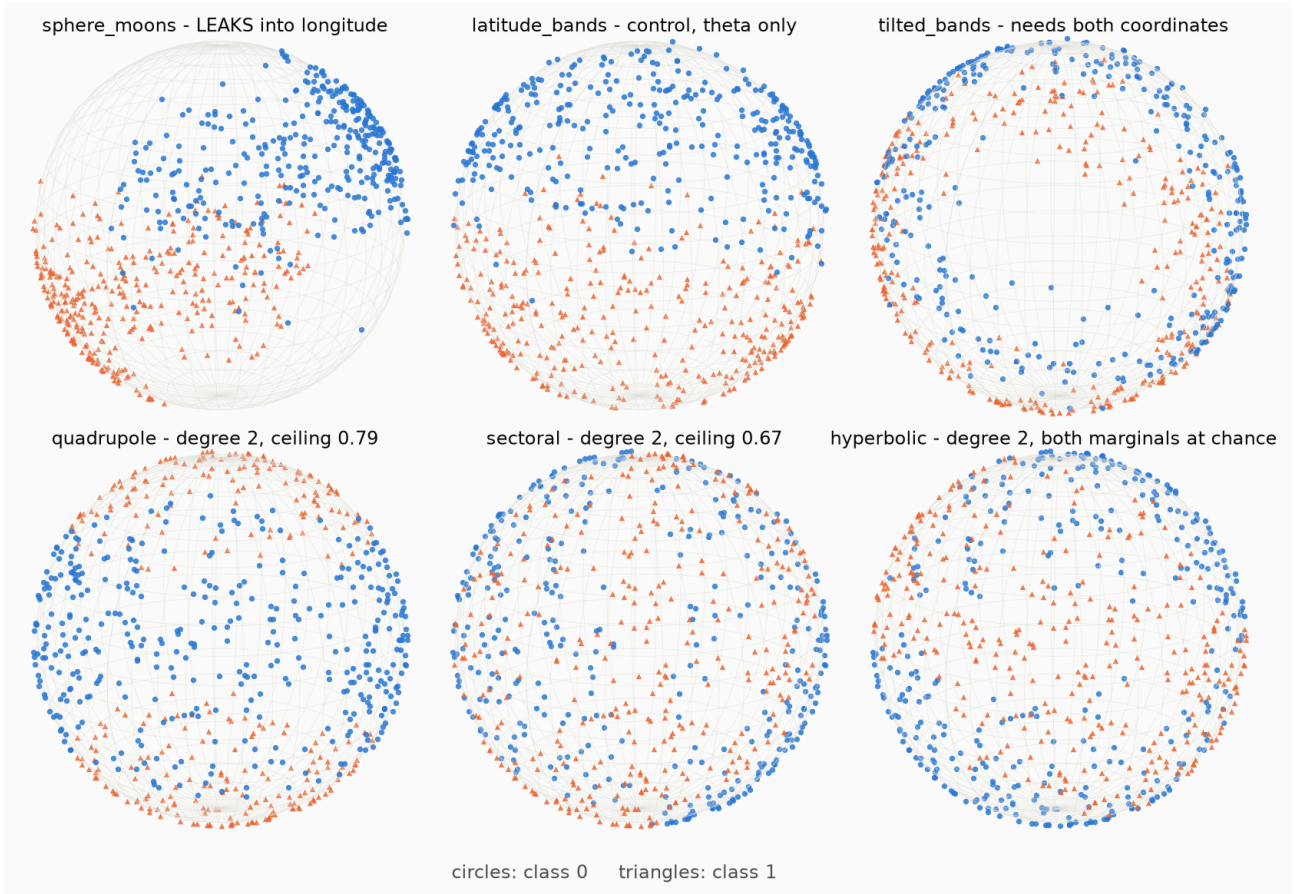


Figure 4. Each dataset on its own sphere, the leaking generator first so the flaw sits beside its replacements. In the first panel the classes split cleanly east/west — that is the longitude leak, visible directly. The second is the axis-aligned control, solvable from latitude by design. The third needs both coordinates. The bottom row is the three falsification targets, each of degree 2 and each provably beyond a single circular boundary.

The gate, and a mistake it caught immediately

The check is automated: every registered dataset is run through it, and a run fails if any dataset does not do what its declared role requires. The first version of the criterion was wrong in an instructive way, and the mistake is worth recording because it is easy to repeat.

That first version flagged a dataset as leaking whenever one coordinate reached the affine ceiling. It promptly rejected the quadrupole, the sectoral target and the banded target. But for a θ -only target the best degree-1 approximation is a θ cut, so the two numbers coincide by construction and say nothing whatever about the dataset. The criterion had to become role-dependent: a dataset meant to be solvable is tested differently from one meant to be out of reach. All seven pass their own test now.

Part V — Experiments and results

Every number below was produced by the committed code on pinned versions (PennyLane 0.45.1, NumPy 2.5.2, SciPy 1.18.1, scikit-learn 1.9.0). The analytic stages ran at full scale; the training sweeps ran at three seeds, which is enough to establish the qualitative result but not to settle small differences.

14. What each experiment tests, and what it returned

Stage 0 — the analytic core

What it tests. Whether the derivations of Parts I–II hold for the actual circuit, and what the ceilings are. No training.

Table 1. Affine ceilings, $n = 20000$ points, 4096 directions. The ceiling is the best accuracy any degree-1 circuit can reach; the majority baseline is what guessing the common class gets.

dataset	degree	balance	majority	ceiling
latitude_bands (control)	1	0.500	0.500	0.983
tilted_bands (primary)	1	0.500	0.500	0.887
sphere_moons (deprecated)	1	0.500	0.500	1.000
quadrupole	2	0.425	0.575	0.784
sectoral	2	0.498	0.502	0.669
banded_4	4	0.453	0.547	0.683
hyperbolic	2	0.498	0.502	0.671

The structural result. Probing the circuit at depths $L = 2, 4, 6$ and 10 , with one qubit and with two, the residual after fitting to $\alpha + \beta \cdot \mathbf{r}$ was of order 10^{-15} in every configuration, with energy above degree 1 zero to the same precision. Equation (16) is confirmed numerically, and the fact that ten layers behave identically to two is exactly the statement that the encoding, not the ansatz, sets the limit.

Stage 1 — the kernel spectrum and the degree-2 wall

Spectrum. Measured eigenvalues divided by n : **0.5004, 0.1764, 0.1622, 0.1610**, and nothing else above numerical noise. Predicted by (19): $0.5, 0.1667, 0.1667, 0.1667$. The trace identity holds to 10^{-8} . The spin ladder's ranks came out at exactly $4, 9, 16, 25$ for $n = 1 \dots 4$, matching the predicted $(n + 1)^2$.

The wall. On the quadrupole the circuit saturates at its ceiling and does not pass it. The control rows are the point of the table:

Table 2. The degree-2 wall, quadrupole target, population ceiling 0.783. Test accuracy.

model	parameters	test	reading
BestAffine (computed, not trained)	4	0.780	the ceiling itself
LogisticOnCartesian	4	0.610	same class, optimising log-loss not accuracy
HarmonicRegression(L=1)	4	0.610	degree-1 by construction; cannot do it
HarmonicRegression(L=2)	9	0.930	degree 2 solves it, as the theory says
Majority baseline	0	0.580	the floor every number must clear
LogisticOnSinCos	5	0.990	beats the circuit — see below

READ THE LAST ROW CAREFULLY — IT IS THE HONEST STORY

A five-parameter linear model on sine–cosine features scores 0.990 on a target the quantum circuit cannot exceed 0.78 on. The reason is exact, not accidental: $\sin \theta$ is one of its features, and the label $|z| > 1/\sqrt{3}$ is *identically* the condition $\sin \theta < \sqrt{(2/3)}$. The target is linearly separable in that representation.

And $\sin \theta = \sqrt{(1 - z^2)}$ is not a polynomial in (x, y, z) , so the sine–cosine representation is **not degree-limited**. It is strictly more expressive than the circuit here, while being less faithful.

So the claim the project can defend is not that the quantum encoding is more powerful. It is that the encoding constrains the model to functions respecting the sphere's geometry — and a constraint is a restriction on power, expected to help when the data genuinely live on a sphere and to hurt otherwise. Writing it any other way would be caught in review, and correctly.

Depth inertness. On the degree-1 target, accuracy was 0.987 at $L = 2$, at $L = 4$ and at $L = 6$ alike, to three decimal places. Prediction R1 confirmed.

Stage 2 — the two ladders

Spectral occupancy. Residual energy above each rung's own degree, averaged over random weight draws:

Table 3. Energy outside $\ell \leq \text{rung}$. Zero means exactly degree-limited.

rung	broadcast (spin) ladder	re-uploading ladder
1	5×10^{-16}	7×10^{-16}
2	9×10^{-16}	0.361
3	1×10^{-15}	0.294
4	2×10^{-15}	0.234

Fourteen orders of magnitude separate the two from rung 2 onward. §9 predicts the left column exactly; §10 predicts the right. This is the project's sharpest measured result, and it required no training whatsoever.

Stage 3 — faithfulness

The design. Generate a dataset once, then rotate it by an element of $\text{SO}(3)$ and re-express it in (θ, φ) . Every geodesic distance is preserved exactly, so the task is *identical* at every placement and any variation in accuracy is

purely representational. Faithfulness is then measured as the **variance across placements**, not the mean — a faithful representation is placement-invariant by construction.

Three axes were run separately, because they test different things: **tilt** asks whether a representation can combine coordinates, **polar** asks whether it respects the topology at $\theta = 0$, and **Haar-random** checks that any effect is not an artefact of one special axis.

Table 4. Standard deviation of balanced accuracy across placements, three seeds. Lower is more placement-invariant. Mean accuracy was 0.85–0.87 for every representation on every axis.

representation	tilt axis	polar axis	Haar-random
bloch (the quantum encoding)	0.0012	0.0000	0.0000
cartesian	0.0049	0.0045	0.0055
sincos	0.0079	0.0064	0.0110
raw (θ, φ)	0.0137	0.0232	0.0248

The quantum encoding is **exactly placement-invariant** — which it must be, since §5 proves the kernel is rotation-invariant, so this is theory confirming itself in a measurement rather than a surprise. The ordering bloch < cartesian < sincos < raw holds on all three axes.

Two honest qualifications belong beside that. The mean accuracies sit within about a point of each other, so the effect lives in the variance and not in the mean; a report claiming the quantum encoding is "more accurate" would be overstating. And these are three seeds — a strong signal, not a settled conclusion. Whether sine–cosine separates from Cartesian by a margin that survives ten seeds is still open, and it is the single most valuable remaining experiment.

Part VI — Writing the report

15. What is proved, what is measured, what is open

The distinction matters more than any individual number, because a reader's trust depends on your not blurring it.

Proved — true by derivation, and the measurements only confirm arithmetic

The encoding places each data point at its own location on the Bloch sphere (§3). Its fidelity kernel is $(1 + \cos \gamma)/2$, a function of geodesic distance alone (§4). That kernel's Legendre expansion terminates at degree 1 (§7). The one-upload model's output is $\alpha + \beta \cdot \mathbf{r}$ exactly — four parameters, any depth, any ancillas — so its decision boundary is a single circle (§7). The Gram matrix therefore has rank at most 4 (§8). The broadcast ladder's kernel is $\cos^{2n}(\gamma/2)$ and its outputs lie exactly within degree n (§9).

These are theorems. Present them as derivations, with the measurements as confirmation that the code implements the mathematics.

Measured — empirical, could have come out otherwise

That re-uploading is not degree-limited (Table 3) follows from the counterexample in §10, but the *size* of the effect is measured. That the affine ceiling on the quadrupole is 0.784 is arithmetic plus sampling. That the circuit saturates at its ceiling rather than falling short of it is a genuinely empirical claim about trainability. That the quantum encoding is exactly placement-invariant while raw coordinates are not (Table 4) is measured, though half-predicted.

Open — do not write as though settled

Whether faithfulness measurably improves *learning*, as opposed to placement-invariance, is not settled. Whether sine–cosine — the standard practitioner's fix — is separable from Cartesian at all is open. Whether the observed variance ordering survives ten seeds is open. Whether any of this transfers to data that is not synthetic is untouched by this work.

THE FRAMING THAT WILL SURVIVE REVIEW

Never write "quantum outperforms classical". The defensible sentence is: *the encoding supplies a prior that the standard convenient featurisations do not* — and after Table 2, be prepared to narrow even that to the raw coordinate chart specifically, unless the ten-seed run separates sine–cosine from Cartesian.

State on the first page that no quantum advantage is claimed, that the model is classically simulable, and that a four-parameter classical model reproduces it on every task within its range. Owning that is far stronger than being caught by it. It also makes the actual contribution legible: the project is about *computing an inductive bias exactly*, not about speed.

A suggested structure

1. **Introduction.** The claim, the exclusions, the dequantisation — all on page one.
2. **Framework.** §§1–4 of this document: the sphere, the geodesic angle, the encoding, and the one-line kernel derivation that unifies them.

3. **The two properties.** §§5–8. Faithfulness is which basis; restriction is how many terms. Give the affine argument, since it explains why depth is irrelevant, which readers find surprising.
4. **Method.** Part IV. Lead with the affine ceiling — a provable bound obtained without training is the methodological contribution, and it is what lets you make falsifiable numerical predictions.
5. **Results.** Part V, in this order: the spectrum, the wall with all its control rows, the ladder contrast, then faithfulness.
6. **What was corrected.** The six items in `CORRECTIONS.md`. Including the two that were your own errors strengthens rather than weakens the work.
7. **Discussion.** The two ladders as distinct knobs. What the standard generalisation bounds cannot see. The open questions above, stated as open.

CHECK YOURSELF — THE READINESS TEST

Before writing, see whether you can do these from memory. Derive $k(\gamma) = (1 + \cos \gamma)/2$ from $\rho = \frac{1}{2}(I + \mathbf{r} \cdot \boldsymbol{\sigma})$. Explain in two sentences why depth does not increase expressivity. State why the Gram matrix has rank 4 without mentioning harmonics. Explain why $\sin \theta \cdot \cos 2\phi$ is not band-limited in degree. Say what the affine ceiling is and why it cannot be blamed on an optimiser. And give the one-sentence honest framing of what the encoding buys.

Six answers. If you have them, the report will write itself — and more importantly, you will be able to defend it.

A closing note on the shape of the result

It is worth saying plainly that the strongest parts of this project are the parts that involve no training at all: a kernel derived in four lines, an expressivity limit proved by linearity, a rank bound from a two-term matrix decomposition, a ceiling computed by sorting, and a degree measurement by least squares. Together they produce falsifiable numerical predictions before a single model is fitted.

That is unusual in machine learning, and it is the thing most worth foregrounding. The work's value does not rest on whether the quantum encoding wins a benchmark. It rests on the encoding's inductive bias being *computable exactly* — which means it can be predicted, checked, and falsified, rather than argued about.